

Logistic models most frequently describe microbial growth curves across contexts

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Abstract

Microbial growth curves are a familiar empirical pattern, but the most useful mathematical description is not always obvious because different models can fit similar trajectories for different reasons. Four candidate models for microbial abundance through time were compared: two phenomenological polynomials (quadratic and cubic) and two sigmoid growth models with interpretable structure (logistic and Gompertz). Each growth curve was defined as a unique Species–Temperature–Medium–Citation–Replicate combination. Models were fitted by least squares and the best-supported model within each curve was selected using AIC and BIC. Across 305 curves, 1178 of 1220 curve–model combinations were successfully fitted and 42 were skipped when parameters were not identifiable from the available time points. A best-supported model could be identified for 297 curves. The logistic model was selected most frequently (AIC: 124/297; BIC: 127/297), followed by the cubic polynomial (AIC: 81/297; BIC: 74/297) and the Gompertz model (68/297 under both criteria). Model discrimination was often weak: 143/297 curves had $\Delta\text{AIC}_{\text{runner-up}} < 2$. Logistic wins were particularly often marginal (61/124 weak wins), whereas cubic wins were more frequently decisive (33/81 with $\Delta\text{AIC}_{\text{runner-up}} > 10$). Overall, a simple logistic form most often captured the dominant structure of microbial growth in this dataset, while the flexibility of cubic polynomials and the asymmetry of the Gompertz curve were advantageous for meaningful subsets of curves.

Introduction

Population growth is conceptually simple but methodologically rich: the same qualitative trajectory can be represented by multiple mathematical forms, each encoding different assumptions and degrees of mechanistic interpretation. Microbes grown in batch culture commonly show a lag phase, a period of approximately exponential increase, and a saturation phase as resources become limiting and inhibitory by-products accumulate (Micha and Corradini, 2011; Zwietering et al., 1990). The resulting curve is typically sigmoidal, but different sigmoids imply different biological emphases. Logistic growth is a compact representation of density dependence and a carrying capacity, whereas the Gompertz curve allows asymmetric rise and saturation and has a long history of use in quantitative microbiology (Buchanan et al., 1997; Zwietering et al., 1990). In parallel, flexible phenomenological functions (such as polynomials) can describe curvature without attributing meaning to parameters; they can therefore fit well but risk over-interpreting noise and can behave poorly outside the observed range (Micha and Corradini, 2011).

When multiple models are plausible, model comparison is best framed as an explicit inferential problem rather than a purely visual choice. Information criteria provide a widely used framework for balancing goodness-of-fit against parameter count (Akaike, 1974; Schwarz, 1978). In ecological and evolutionary applications, this approach has become standard practice for comparing alternative hypotheses represented by alternative models (Burnham and Anderson, 2002; Johnson and Omland, 2004). However, nonlinear fitting can be fragile: convergence, starting values, bounds, and data resolution can all influence whether parameter estimation succeeds and whether models are distinguishable (Bolker et al., 2013; Motulsky and Christopoulos, 2004).

The present analysis addresses a focused question: *Across microbial growth curves measured under diverse experimental contexts, which mathematical model is most often best supported by the data?* Two polynomial benchmarks (quadratic and cubic) are compared with two widely used sigmoid growth models (logistic and Gompertz). In addition, a fully reproducible workflow is implemented that starts from raw data, produces fitted models and summary figures, and compiles a report from a single command.

Methods

Data

The course-provided dataset `logistic_growth_data.csv` was used. It contains time-series measurements of microbial abundance or biomass (`PopBio`) through time (`Time`) across species/strains, temperatures, media, citations and experimental replicates. Time was recorded in hours. `PopBio` was recorded in heterogeneous units (e.g. optical density, CFU, counts, dry weight). Because measurement scale and error structure differ across units, model selection was performed *within* each curve (models compared only on the same response scale), rather than by comparing residual magnitudes across curves (Burnham and Anderson, 2002; Johnson and Omland, 2004).

A single growth curve was defined by the unique combination of `Species`, `Temp`, `Medium`, `Citation`, and `Rep`. These fields were concatenated to construct a unique identifier (`curve_id`) and a shorter numeric index (`curve_num`) for plotting and tabulation. Within each curve, ob-

servations were ordered by time prior to fitting. Minor floating-point artefacts around Time = 0 were treated as zero for plotting only.

Candidate models

For each curve, four candidate models describing PopBio as a function of time t were fitted:

1. **Quadratic polynomial** (phenomenological): $y(t) = at^2 + bt + c$.
2. **Cubic polynomial** (phenomenological): $y(t) = at^3 + bt^2 + ct + d$.
3. **Logistic growth** (sigmoid): $y(t) = \frac{K}{1 + \exp\{-r(t - t_0)\}}$, where K is an upper asymptote (carrying-capacity proxy), r is a growth-rate parameter, and t_0 is an inflection-time parameter.
4. **Gompertz growth** (asymmetric sigmoid): $y(t) = K \exp\{-\exp[-r(t - t_0)]\}$, where K is the upper asymptote, r controls the maximum growth rate near the inflection, and t_0 is a timing parameter (Buchanan et al., 1997; Zwietering et al., 1990).

Polynomial models provide flexible benchmarks but lack mechanistic interpretation and can extrapolate poorly outside the observed range (Micha and Corradini, 2011). Logistic and Gompertz models are widely used primary growth models in microbiology and summarise multi-phase trajectories with interpretable parameters (Micha and Corradini, 2011; Zwietering et al., 1990). In this report, logistic and Gompertz are treated as “mechanistic-inspired” rather than fully mechanistic: expected qualitative structure is captured and parameters remain interpretable, but explicit resource dynamics and physiological lag processes are abstracted away.

Model fitting

All models were fitted by least squares using `scipy.optimize.curve_fit`. For polynomial models, coefficients from `numpy.polyfit` were used as starting values. For sigmoid models, starting values were derived from each curve: K was initialised to the maximum observed PopBio (or slightly above), t_0 to the median observed time, and r to a small positive value. To stabilise estimation across heterogeneous curves, bounds reflecting basic biological plausibility were imposed: $K > 0$, $r > 0$, and t_0 constrained to fall within the observed time range. This bounded NLLS approach improves stability for heterogeneous datasets and reduces implausible solutions (Bolker et al., 2013; Motulsky and Christopoulos, 2004).

Some curve–model combinations were **skipped** when parameters were not identifiable from the available time points. Specifically, if a curve had fewer than $k + 1$ unique time values (where k is the number of free parameters in the model), the fit was not attempted. Skipped fits were treated as data-resolution limitations rather than numerical failures (Bolker et al., 2013).

Model selection and evidence strength

For each fitted curve–model combination, residual sum of squares (RSS) was calculated and used to compute AIC and BIC under a Gaussian error model with constant variance. With

99 n observations and k fitted parameters, the following standard least-squares forms were used
 100 (Akaike, 1974; Schwarz, 1978):

$$\text{AIC} = n \ln \left(\frac{\text{RSS}}{n} \right) + 2k, \quad \text{BIC} = n \ln \left(\frac{\text{RSS}}{n} \right) + k \ln(n).$$

101 Models were compared only within the same curve. The winning model for each curve was the
 102 one with minimum AIC (or BIC). To quantify how decisively the winner was preferred, the AIC
 103 gap between the best and second-best model within each curve was computed as $\Delta\text{AIC}_{\text{runner-up}}$,
 104 and evidence was summarised using conventional thresholds (< 2 , $2-7$, $7-10$, > 10) (Burnham
 105 and Anderson, 2002). Representative fitted curves were inspected visually to ensure that low
 106 information criteria corresponded to sensible fitted shapes rather than pathological parameteri-
 107 sations.

108 Computing tools

109 All analysis steps were implemented in Python and executed from a single entry point (`run_MiniProject.py`),
 110 which runs data preparation, exploratory plotting, model fitting, model comparison, and sum-
 111 mary steps sequentially. Data handling used `pandas`; numerical operations used `numpy`; plotting
 112 used `matplotlib`; and model fitting used `scipy`. Git was used for version control. The report
 113 was written in L^AT_EX and compiled via a shell script to make document generation reproducible.

114Results

115Data overview and fitting coverage

116 A total of 305 unique growth curves were identified, and four models were attempted per curve
 117 (1220 curve–model combinations). Model fitting succeeded for 1178 combinations, and 42 com-
 118 binations were skipped due to insufficient unique time points for parameter identifiability. A
 119 best-supported model (AIC/BIC computable and comparable) could be identified for 297 curves.
 120 Seven curves contained at least one non-positive `PopBio` value, consistent with measurement
 121 noise or background-correction artefacts, which restricts the use of naive log transformations
 122 without additional handling.

123Overall model performance

124 Under AIC, the logistic model was selected most frequently (124/297 curves; 41.8%), followed by
 125 the cubic polynomial (81/297; 27.3%), the Gompertz model (68/297; 22.9%), and the quadratic
 126 polynomial (24/297; 8.1%). The pattern was similar under BIC, which slightly increased support
 127 for the logistic model (127/297; 42.8%) and reduced support for the cubic polynomial (74/297;
 128 24.9%) (Table 1; Figure 1). Overall, the ranking logistic $>$ cubic $>$ Gompertz $>$ quadratic was
 129 consistent across both criteria.

Table 1: Model win counts under AIC and BIC (n=297 curves with an identifiable winner).

Model	AIC wins	BIC wins	Total (AIC)	Total (BIC)
Logistic	124	127	297	297
Cubic	81	74	297	297
Gompertz	68	68	297	297
Quadratic	24	28	297	297

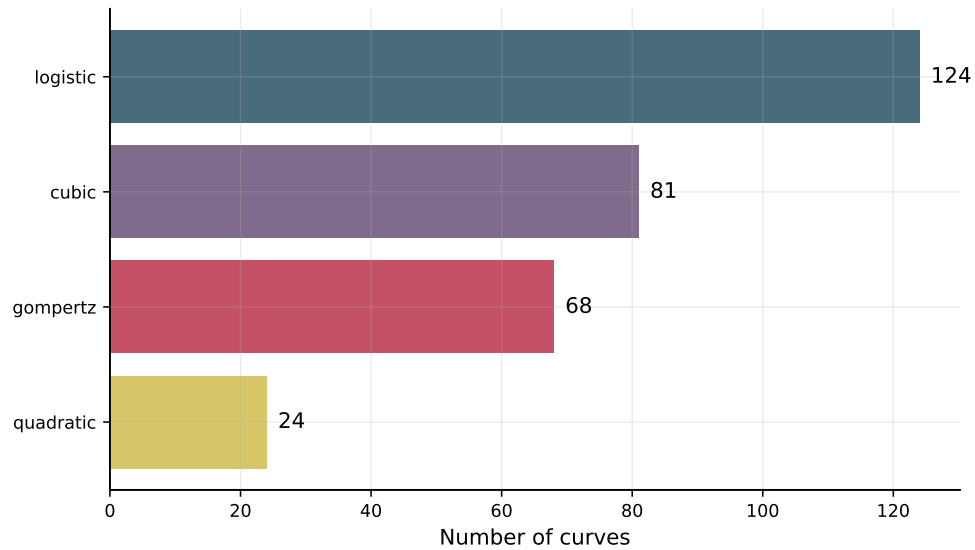


Figure 1: **Overall model winners across growth curves under AIC.** Bars show how often each model achieved the lowest AIC within a growth curve (n=297 curves with an identifiable winner). Logistic growth was selected most often, with cubic polynomials and the Gompertz model providing better support for substantial subsets of curves.

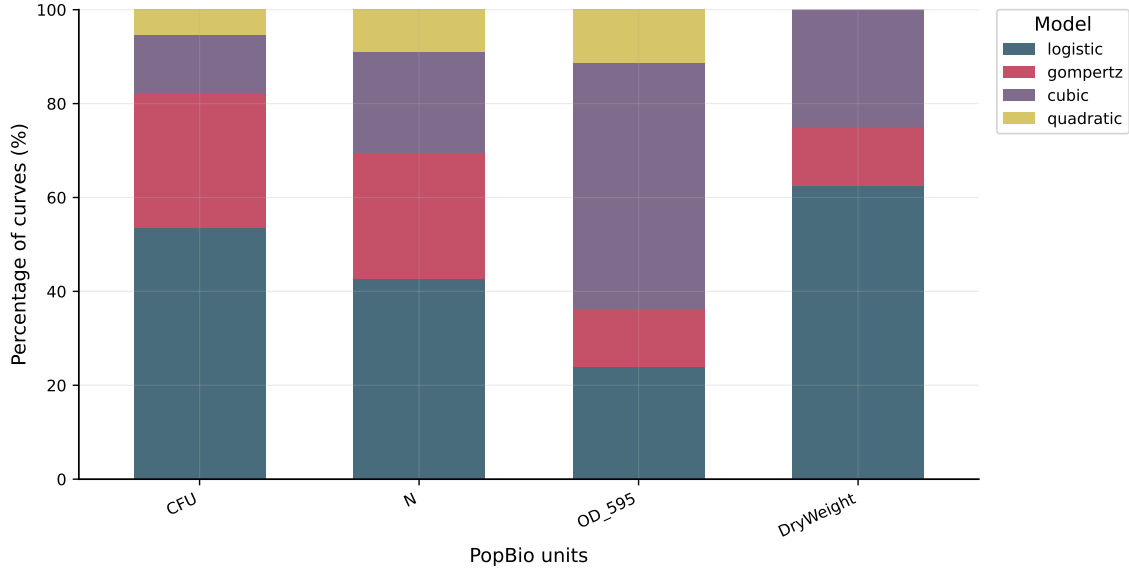


Figure 2: **AIC-winning models stratified by response units.** Winners are shown separately for different PopBio measurement scales. The relative frequency of polynomial vs. sigmoid winners varies among units, consistent with heterogeneous curve shapes and noise structures across experimental contexts.

Evidence strength of model selection

Win counts alone can obscure whether the winning model is clearly preferred or only marginally better than its closest competitor. Across 297 curves, discrimination was frequently weak: 143 curves (48.1%) had $\Delta\text{AIC}_{\text{runner-up}} < 2$, indicating that multiple models fit nearly equally well. A further 63 curves (21.2%) showed moderate support (2–7), 31 curves (10.4%) strong support (7–10), and 60 curves (20.2%) very strong support (> 10) for the winner (Burnham and Anderson, 2002).

Evidence strength differed among winners. Logistic wins were often marginal: 61/124 (49.2%) of logistic-winning curves had $\Delta\text{AIC}_{\text{runner-up}} < 2$. Gompertz wins were similarly often weak (35/68, 51.5% weak) and only rarely very strong (4/68, 5.9%). In contrast, cubic wins were more frequently decisive: 33/81 (40.7%) of cubic-winning curves had $\Delta\text{AIC}_{\text{runner-up}} > 10$. This pattern suggests that many curves do not contain sufficient information to confidently discriminate among similar sigmoids, while a subset contain departures from simple sigmoid structure that are captured much better by more flexible functions (Figure 4).

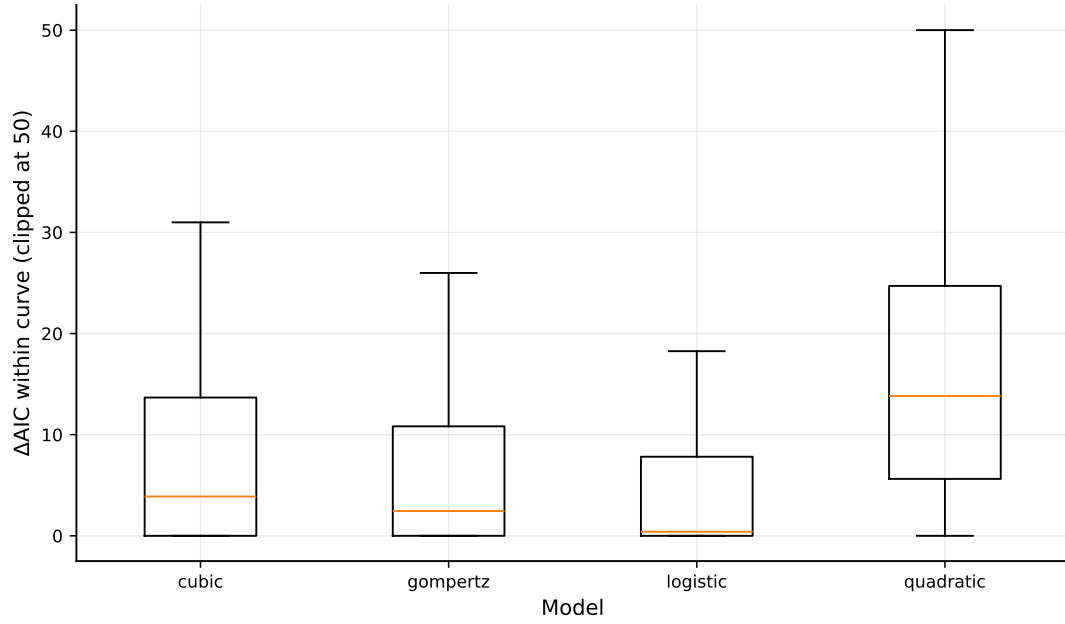


Figure 3: **Within-curve Δ AIC distributions across models.** For each growth curve, Δ AIC is computed as AIC minus the minimum AIC among the four candidate models for that curve. Values close to 0 indicate near-best fits, whereas larger values indicate poorer support within that curve.

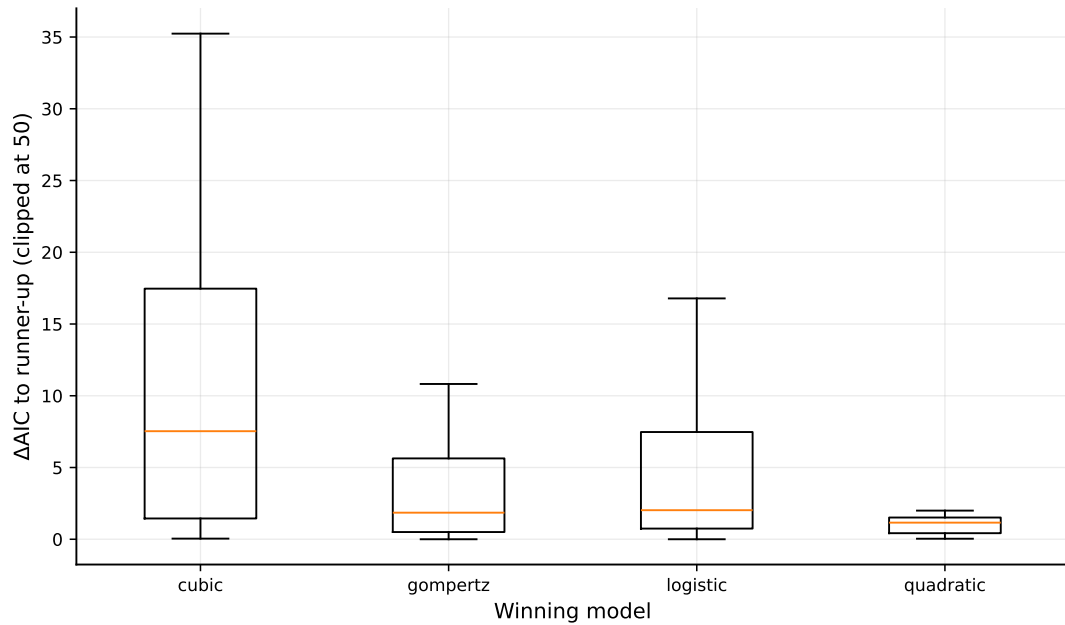


Figure 4: **Evidence strength of AIC winners.** For each curve, the AIC gap between the best and second-best model (Δ AIC_{runner-up}) summarises how decisively the winner is preferred. Small gaps (< 2) indicate weak discrimination among models; large gaps (> 10) indicate strong support for the winner.

When does each model win? Links to sampling density and curve scale

Model winners were not evenly distributed across curve informativeness. The number of observations per curve influenced which model was selected (Table 2; Figure 5). For sparsely sampled curves (≤ 10 points), the logistic model won most often, consistent with its ability to summarise overall sigmoid structure with few parameters. As sampling density increased (20–30 points), cubic wins became relatively more common, consistent with better-resolved curvature providing the information needed to justify additional flexibility.

Winner patterns also depended on within-curve variation (dynamic range; Table 3; Figure 6). In the lowest dynamic-range quartile, cubic polynomials dominated, whereas logistic wins increased with dynamic range. When a curve does not span a large range of PopBio values, both rapid growth and saturation may be poorly resolved; in that setting, polynomials can fit shallow curvature without requiring a well-identified plateau. Conversely, when dynamic range is large, saturation becomes more detectable and logistic structure is more strongly supported. Gompertz winners were concentrated in the highest dynamic-range quartile, consistent with asymmetry becoming detectable only when curves span sufficient range and are sampled densely enough.

Table 2: AIC winners stratified by sampling density (number of observations per curve).

n points bin	Cubic	Gompertz	Logistic	Quadratic
≤ 10	29	25	62	4
10–15	18	25	38	9
15–20	12	11	12	4
20–30	21	3	10	7
≥ 30	1	3	1	0

Table 3: AIC winners stratified by within-curve PopBio dynamic-range quartiles.

Dynamic-range quartile	Cubic	Gompertz	Logistic	Quadratic
Q1 (lowest)	39	5	16	15
Q2	22	26	26	0
Q3	16	13	44	1
Q4 (highest)	4	24	38	8

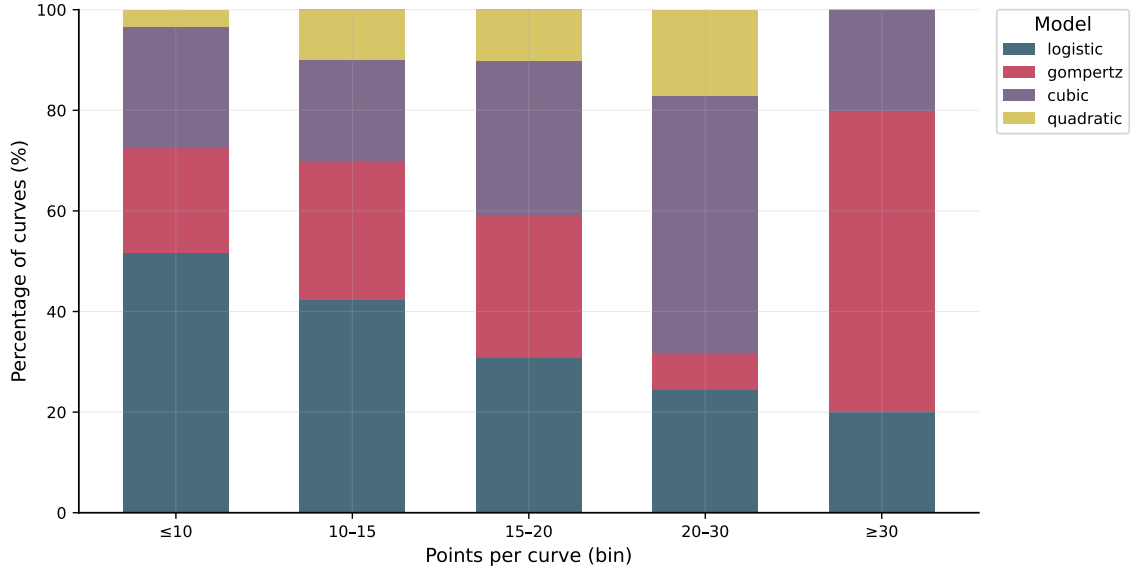


Figure 5: **AIC-winning model composition stratified by sampling density.** Bars show the percentage of curves (within each bin of number of observations) for which each model is the AIC winner. Better-sampled curves show a higher fraction of polynomial wins, consistent with additional curvature becoming detectable.

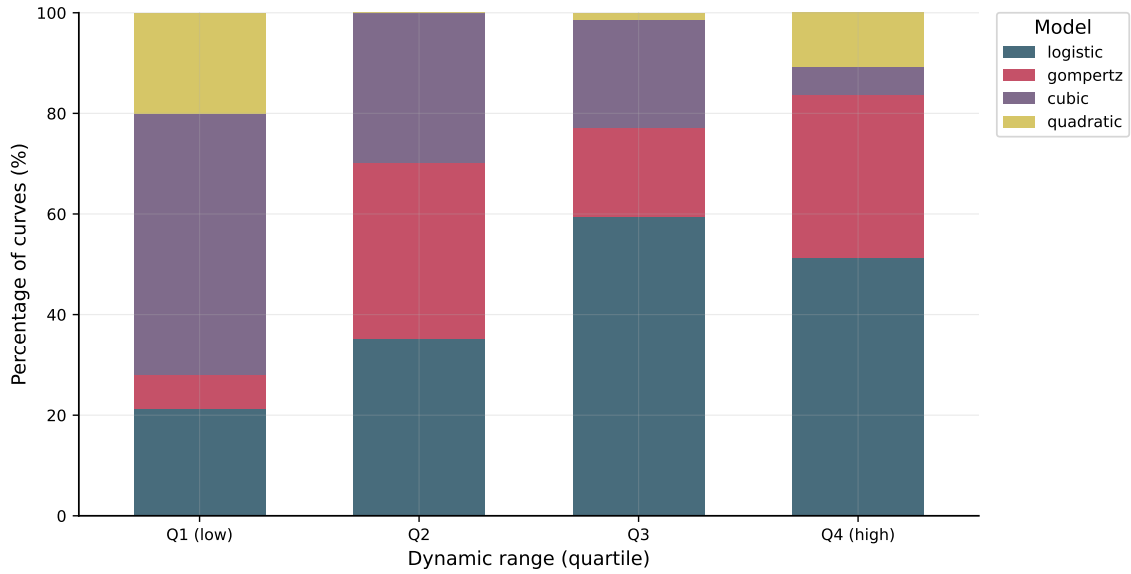


Figure 6: **AIC-winning model composition stratified by dynamic range.** Bars show the percentage of curves (within each dynamic-range quartile) for which each model is the AIC winner. Logistic wins increase with dynamic range, whereas cubic wins dominate low dynamic-range curves, consistent with weakly informative trajectories where sigmoid structure is hard to distinguish.

Discussion

Across diverse microbial growth curves, the logistic model was most frequently best supported under both AIC and BIC. Many trajectories exhibit a clear approach to a plateau, and logistic growth summarises that structure parsimoniously. BIC in particular rewards models that capture dominant patterns without unnecessary flexibility (Raftery, 1995; Schwarz, 1978). The Gompertz model, although selected less often overall, still won for nearly a quarter of curves, suggesting that asymmetric rise versus saturation is common but not universal in this dataset, consistent with previous comparisons of primary growth models (Buchanan et al., 1997; Zwietering et al., 1990).

A key insight from the ΔAIC analysis is that “winning” often reflects marginal differences rather than decisive evidence: almost half of curves had $\Delta\text{AIC}_{\text{runner-up}} < 2$. In such cases, multiple model classes can reproduce the observed data equally well, either because curves are sparsely sampled, because measurement noise dominates subtle shape differences, or because curves do not span enough dynamic range to clearly reveal both rapid growth and saturation. When discrimination is weak, model choice is often better guided by interpretability and intended use than by small information-criterion differences (Burnham and Anderson, 2002; Johnson and Omland, 2004). This is particularly relevant for logistic winners: nearly half were weak wins, implying that logistic growth is frequently a reasonable default summary but not uniquely supported by the data.

Conversely, the relatively high frequency of very strong cubic wins indicates that a subset of curves contain structure that simple sigmoids do not capture. In microbial growth data this can arise from late-stage curvature changes, mild declines after a peak, or measurement artefacts; flexible polynomials can reduce residual error dramatically, but parameters are not biologically interpretable and extrapolation can be unstable (Micha and Corradini, 2011). These results highlight a recurring modelling trade-off: mechanistic-inspired simplicity yields interpretable summaries, while flexibility can be decisive when empirical trajectories depart from idealised assumptions.

The stratified winner analysis helps explain these patterns. Logistic wins increased with dynamic range, consistent with saturation being most detectable when curves span a large range of PopBio. Cubic wins dominated low dynamic-range curves, where sigmoid structure is hard to identify and where the data may not constrain carrying-capacity-like parameters. Gompertz winners were concentrated in the highest dynamic-range quartile, suggesting that asymmetry is easiest to detect when the curve spans enough range and is sampled densely enough to separate subtle sigmoid shapes. These patterns align with general guidance for nonlinear modelling: identifiability depends not only on sample size but on where samples fall relative to informative regions of the curve (Bolker et al., 2013; Motulsky and Christopoulos, 2004).

Several limitations shape how far these results can be generalised. First, PopBio is measured in heterogeneous units, implying different error structures and potential heteroskedasticity; inference was therefore restricted to within-curve comparisons. Second, log-scale approaches that can stabilise variance are not universally applicable: seven curves contained non-positive PopBio values, making naive log transformation invalid without additional handling. Third, skipped fits reflect genuine data-resolution constraints and indicate that model complexity must match curve

informativeness. Finally, logistic and Gompertz models compress multi-phase microbial growth into simple sigmoids; models that explicitly represent lag-phase dynamics (e.g. Baranyi-type approaches) could be valuable extensions if the goal were physiological inference about lag rather than comparative description (Baranyi and Roberts, 1994; Buchanan et al., 1997).

Overall, the central conclusion is robust: across this dataset, logistic growth most often captures the dominant empirical structure of microbial growth curves, but model discrimination is frequently weak and subsets of curves show strong support for alternative shapes. This supports reporting not only which model wins most often, but also how decisive those wins are, rather than relying on winner counts alone.

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